The University of Sydney

ELEC5305 Project

Does Wiener Filtering Help Modern Speech Foundation Models?

A Layer-Wise Analysis

Final Project Report

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GitHub Project Site

`https://jennywu55.github.io/elec5305-project-550203623/`

GitHub Repository

`https://github.com/JennyWu55/elec5305-project-550203623`

Project Demonstration Video

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Abstract

This project investigates whether classical STFT-based Wiener speech enhancement improves not only waveform quality but also the internal representations of modern self-supervised speech models. Four LibriSpeech utterances were mixed with DKITCHEN and STRAFFIC noise from the DEMAND database at input SNRs of -5, 0, 5 and 10 dB, producing 32 controlled conditions. A fixed Wiener enhancement system was evaluated using SNR improvement, Short-Time Objective Intelligibility (STOI), and layer-wise cosine similarity in WavLM Base+ and HuBERT Base.

Wiener filtering improved waveform SNR by an average of 7.27 dB for DKITCHEN and 4.64 dB for STRAFFIC, while STOI changed only slightly. Representation recovery was strongest in early transformer layers and generally decreased with depth. HuBERT recovery decreased from 0.0551 at layer 1 to 0.0043 at layer 12, while WavLM retained 0.0178 at its final layer. The results show that better waveform SNR does not always lead to better high-level neural representations, and that the value of external enhancement depends on noise level, model and network depth.

1 Introduction

Speech recorded in real environments is often affected by traffic, machinery, ventilation systems, conversations and other background sounds. Speech enhancement attempts to reduce this interference while preserving the useful speech signal.

Wiener filtering is a classical method for noise reduction. In speech processing, it can be applied in the short-time frequency domain by using the Short-Time Fourier Transform (STFT), estimating the noise spectrum, applying a frequency-dependent gain, and reconstructing the enhanced signal.

Traditional speech enhancement is usually evaluated using waveform or perceptual measures such as SNR and intelligibility. Modern speech systems, however, increasingly use self-supervised models such as HuBERT [1] and WavLM [2]. These models convert speech into learned hidden representations across multiple transformer layers.

This creates a useful question. A denoiser may improve the waveform, but does the processed signal also become more similar to clean speech inside a modern speech model? It is possible for noise reduction to help the model, but it is also possible for processing artefacts to change information used by deeper network layers.

1.1 Research Question

The research question is:

> *Does classical Wiener speech enhancement make noisy speech representations more similar to clean speech inside modern self-supervised speech foundation models, and how does this effect vary with model depth, noise type and input SNR?*

1.2 Project Contributions

The project:

- implements an STFT-based Wiener speech enhancement pipeline;
- evaluates it using two real DEMAND noise environments and four controlled SNR levels;

- compares SNR, STOI and foundation-model representation similarity;
- analyses WavLM Base+ and HuBERT Base at layers 1, 4, 8 and 12;
- measures whether Wiener enhancement moves noisy representations closer to clean speech.

2 Literature Review

2.1 Wiener Speech Enhancement

The Wiener filter estimates a desired signal from an observation contaminated by additive noise. It is based on minimum mean-square-error estimation [3].

For short-time speech processing, a noisy STFT coefficient can be written as

$$Y(k,m) = S(k,m) + N(k,m), \tag{1}$$

where $S(k,m)$ is clean speech and $N(k,m)$ is noise.

A Wiener gain can be expressed as

$$G(k,m) = \frac{P_S(k,m)}{P_S(k,m) + P_N(k,m)}, \tag{2}$$

where P_S and P_N are speech and noise power estimates.

The enhanced spectrum is then

$$\hat{S}(k,m) = G(k,m)Y(k,m). \tag{3}$$

The method is simple and computationally efficient, but its quality depends on the noise estimate. Strong suppression can reduce noise while also changing parts of the speech spectrum. This noise-distortion trade-off is a common issue in speech enhancement [4].

2.2 Speech Intelligibility

SNR measures the amount of waveform error but does not directly measure intelligibility. STOI was developed to estimate intelligibility by comparing short-time speech information between clean and degraded signals [5].

Using both SNR and STOI therefore provides two different views of the same enhancement system: one focused on signal error and one more closely related to speech intelligibility.

2.3 Self-Supervised Speech Models

Self-supervised learning allows speech models to learn representations from large amounts of audio without requiring full manual transcription. wav2vec 2.0 [6], HuBERT [1], and WavLM [2] are examples of this approach.

HuBERT learns by predicting masked hidden units derived from speech. WavLM uses a related self-supervised framework and was designed for a wide range of speech-processing tasks.

Both models contain multiple transformer layers. The representation therefore changes as speech moves through the network. This makes layer-wise analysis useful when studying the effect of external speech enhancement.

2.4 Research Gap

The ability of Wiener filtering to reduce additive noise is already well known. The focus of this project is therefore not whether Wiener filtering can increase SNR.

Instead, the project investigates whether the same enhancement also helps representations inside modern self-supervised speech models. This connects a classical signal-processing method with the type of learned features increasingly used in current speech systems.

3 Methodology

3.1 Clean Speech

Clean speech was selected from the LibriSpeech `dev-clean` dataset [7].

Four utterances from two speakers were used:

- 84-121123-0014;
- 84-121123-0015;
- 1988-24833-0021;
- 1988-24833-0022.

All signals were processed at 16 kHz.

3.2 Environmental Noise

Environmental noise was taken from the DEMAND database [8]. Two environments were selected:

- **DKITCHEN**: indoor kitchen noise;
- **STRAFFIC**: street-traffic noise.

One channel from each DEMAND recording was used. The two noises were chosen because they have different acoustic behaviour and therefore provide a more useful test than a single synthetic noise source.

3.3 Controlled Mixture Generation

Each clean utterance was mixed with both noise types at

$$-5, \quad 0, \quad 5, \quad 10 \text{ dB}. \tag{4}$$

The experiment therefore contained

$$4 \text{ utterances} \times 2 \text{ noise types} \times 4 \text{ SNR levels} = 32 \text{ conditions}. \tag{5}$$

For each utterance/noise pair, the same noise segment was used at every SNR level. Only the noise amplitude was changed. This kept the SNR comparison controlled.

The noise scale was calculated from measured clean-speech and noise powers. The generated mixtures were checked to confirm that measured input SNR matched the requested SNR.

A one-second noise-only region was added before speech onset and was used to estimate the noise spectrum for Wiener filtering.

3.4 STFT-Based Wiener Enhancement

The same Wiener configuration was used for all 32 conditions.

Table 1: Fixed Wiener enhancement parameters.

Parameter	Value
Sampling frequency	16 kHz
Frame length	512 samples
Hop length	256 samples
FFT length	512
Window	Hann
Gain floor	0.12
Temporal smoothing	0.70

Noise power spectral density was estimated from the initial noise-only frames.

Speech power was estimated as

$$\hat{P}_S(k,m) = \max\left(|Y(k,m)|^2 - P_N(k), 0\right). \tag{6}$$

A Wiener gain was then calculated using the estimated speech and noise powers.

A gain floor of 0.12 was used to avoid excessive attenuation. Temporal smoothing was applied as

$$\tilde{G}_m = 0.70\tilde{G}_{m-1} + 0.30G_m. \tag{7}$$

The enhanced waveform was reconstructed with inverse FFT processing and weighted overlap-add.

3.5 Signal-Level Evaluation

For clean speech $s[n]$ and an estimated signal $\hat{s}[n]$, SNR was

$$\mathrm{SNR} = 10\log_{10}\left(\frac{\sum_n s[n]^2}{\sum_n (s[n]-\hat{s}[n])^2}\right). \tag{8}$$

SNR improvement was

$$\Delta\mathrm{SNR} = \mathrm{SNR}_{\mathrm{output}} - \mathrm{SNR}_{\mathrm{input}}. \tag{9}$$

STOI was calculated between clean speech and both the noisy and Wiener-enhanced signals. STOI improvement was

$$\Delta\mathrm{STOI} = \mathrm{STOI}_{\mathrm{Wiener}} - \mathrm{STOI}_{\mathrm{noisy}}. \tag{10}$$

3.6 Foundation-Model Evaluation

Two pretrained models were used:

- WavLM Base+;
- HuBERT Base.

The models were used without fine-tuning.

Clean, noisy and Wiener-enhanced versions of each utterance were passed through the same model. Hidden representations were extracted at transformer layers 1, 4, 8 and 12.

These layers provide samples from early, lower-middle, upper-middle and final stages of the networks.

3.7 Representation Recovery Score

Within each model and layer, cosine similarity was calculated between clean and noisy representations,

$$S_{\mathrm{noisy}} = \cos\left(H_{\mathrm{clean}}, H_{\mathrm{noisy}}\right), \tag{11}$$

and between clean and Wiener-enhanced representations,

$$S_{\mathrm{Wiener}} = \cos\left(H_{\mathrm{clean}}, H_{\mathrm{Wiener}}\right). \tag{12}$$

Frame-wise similarity was averaged across time for each utterance.

The representation recovery score was defined as

$$R = S_{\mathrm{Wiener}} - S_{\mathrm{noisy}}. \tag{13}$$

A positive value means that Wiener enhancement moved the representation closer to clean speech. A negative value means that it moved further away.

Results were averaged across the four utterances for each experimental condition.

4 Results

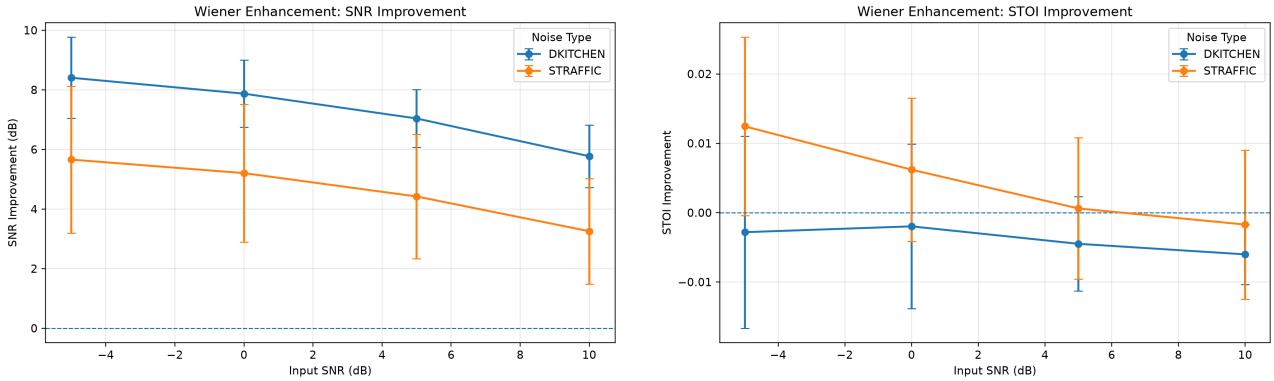
4.1 Signal-Level Results

Wiener enhancement increased SNR in every tested condition.

For DKITCHEN, mean SNR improvement decreased from 8.40 dB at −5 dB input SNR to 5.77 dB at 10 dB input SNR. The average improvement across the four SNR levels was approximately 7.27 dB.

For STRAFFIC, improvement decreased from 5.66 dB at −5 dB to 3.26 dB at 10 dB, with an overall average of approximately 4.64 dB.

STOI showed a different pattern. DKITCHEN produced small negative STOI changes across the tested SNR levels. STRAFFIC produced small positive changes at −5 and 0 dB, almost no change at 5 dB, and a small negative change at 10 dB.



(a) SNR improvement.

(b) STOI improvement.

Figure 1: Signal-level effects of Wiener enhancement under DKITCHEN and STRAFFIC noise. Error bars show standard deviation across four utterances.

Figure 1 shows that large SNR improvement did not produce a similarly large STOI improvement. This is the first indication that the effect of enhancement depends on the metric being measured.

4.2 Layer-Wise Representation Recovery

Both models showed their strongest recovery in early layers.

For WavLM, mean recovery decreased from 0.0454 at layer 1 to 0.0178 at layer 12.

For HuBERT, the decrease was stronger, from 0.0551 at layer 1 to 0.0043 at layer 12.

Table 2: Mean representation recovery across transformer depth.

Model	Layer 1	Layer 4	Layer 8	Layer 12
HuBERT Base	0.0551	0.0266	0.0140	0.0043
WavLM Base+	0.0454	0.0220	0.0242	0.0178

HuBERT produced positive recovery in 100% of conditions at layer 1, decreasing to 50% at layer 12. For WavLM, the corresponding values were 93.75% and 62.50%.

Figure 2 also shows that recovery was generally larger under the lowest input SNR conditions. When the input speech was already relatively clean, Wiener enhancement produced a smaller representation change.

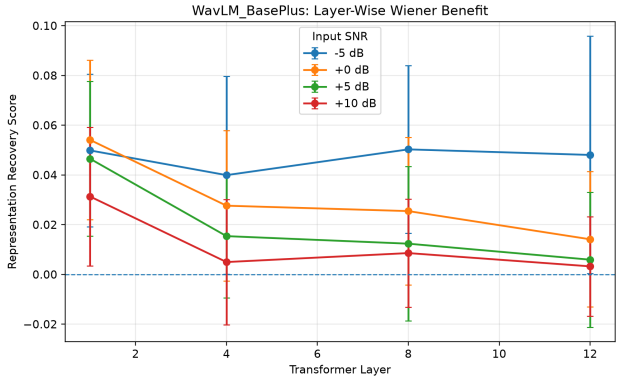
4.3 Model Comparison

HuBERT showed stronger recovery than WavLM in layer 1, but its recovery decreased more quickly with depth.

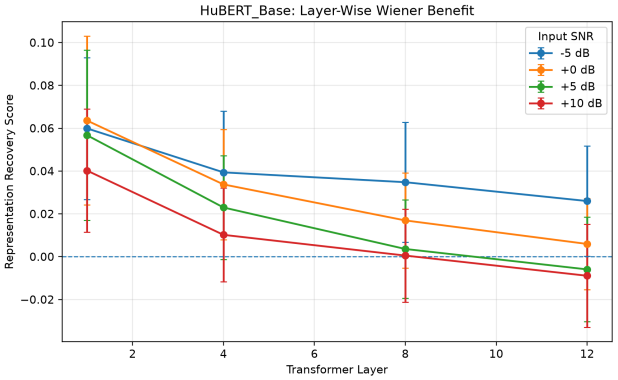
WavLM retained a larger positive mean recovery at its final layer.

Across experiments, the Pearson correlation between SNR improvement and mean representation recovery was approximately 0.57 for HuBERT and 0.69 for WavLM.

The positive correlations show that stronger noise reduction often helped the internal representation. However, the scatter in Figure 3b shows that SNR alone did not fully predict the representation result.

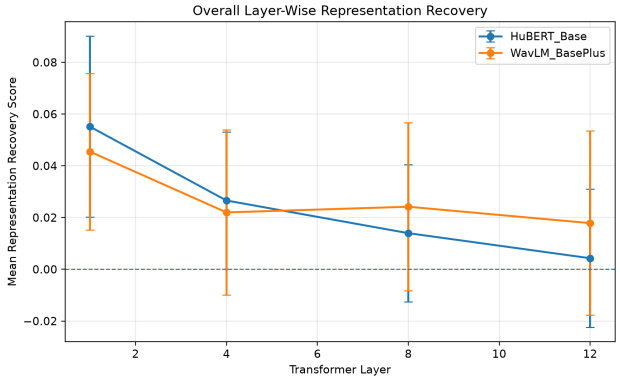


(a) WavLM Base+.

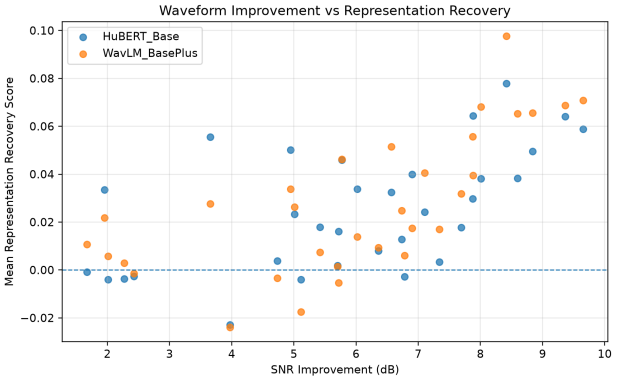


(b) HuBERT Base.

Figure 2: Layer-wise representation recovery under four input SNR conditions. Positive values indicate movement toward the clean-speech representation.



(a) Overall layer-wise comparison.



(b) SNR gain versus representation recovery.

Figure 3: Comparison of model depth and the relationship between waveform improvement and representation recovery.

4.4 Representative Spectrogram

Figure 4 shows one STRAFFIC condition at 0 dB input SNR.

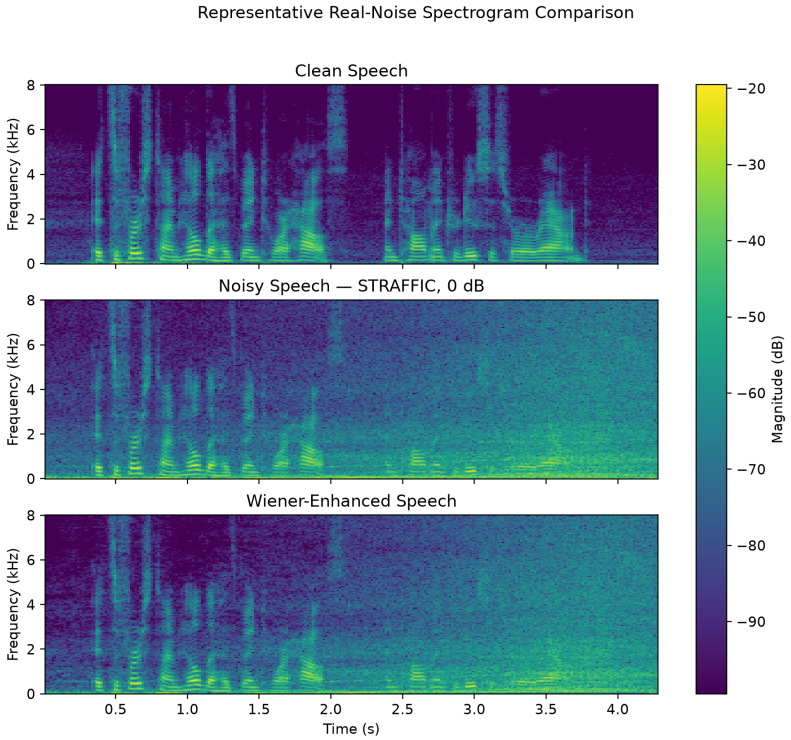


Figure 4: Representative clean, noisy and Wiener-enhanced spectrograms for STRAFFIC at 0 dB.

Compared with the noisy signal, the enhanced spectrogram contains less broadband background energy while retaining the main speech structure. Residual noise remains after enhancement, which is consistent with the smaller SNR improvement observed for STRAFFIC.

5 Discussion

5.1 SNR Improvement and STOI Do Not Tell the Same Story

The most direct result is that the Wiener system successfully reduced waveform error. All 32 experiments produced positive SNR improvement.

The improvement was substantially larger for DKITCHEN than for STRAFFIC. The initial noise estimate appears to work better for DKITCHEN, while traffic noise changes more over time and is harder to represent with a fixed noise spectrum.

The STOI result is more interesting. Even when SNR improved by several decibels, STOI usually changed by only a small amount.

This means that removing noise energy did not automatically improve the speech information measured by STOI. The Wiener filter reduced background energy, but it also changed parts of the speech spectrum. The balance between these two effects was different from the balance measured by SNR.

For this project, the result shows why evaluating enhancement with only one metric would give an incomplete picture.

5.2 Wiener Enhancement Helps Early Representations Most

Layer 1 showed the strongest and most consistent representation recovery in both HuBERT and WavLM.

This result makes sense because early model representations are still closely related to acoustic information in the input waveform. Removing background noise therefore makes the early representation look more like the representation generated from clean speech.

The benefit was also generally larger at low input SNR. When the input contained heavy noise, the difference between noisy and clean speech was large enough for the denoiser to produce a clear representation change.

At higher input SNR, the original speech was already closer to clean speech, so the additional benefit of Wiener filtering became smaller.

5.3 The Benefit Decreases in Deeper Layers

The main finding of the project is that the early-layer benefit did not continue equally through the network.

HuBERT showed the strongest example. Mean recovery decreased from 0.0551 at layer 1 to 0.0043 at layer 12, and only half of the final-layer conditions had positive recovery.

This suggests that deeper HuBERT representations are less directly dependent on the waveform noise removed by Wiener filtering. By the final layer, the network has already transformed the acoustic input through several contextual stages.

Wiener enhancement can also change speech components at the same time as it removes noise. These changes may be unimportant for waveform SNR but still affect the representation produced by a later transformer layer.

The result therefore answers the main research question more clearly than the SNR result alone: external denoising is useful in representation space, but the benefit depends strongly on where the representation is measured.

5.4 WavLM and HuBERT Respond Differently

The two models did not react to Wiener enhancement in exactly the same way.

HuBERT had the larger layer-1 recovery, while WavLM retained more recovery at deeper layers. At layer 12, WavLM had an average recovery of 0.0178 compared with only 0.0043 for HuBERT.

This shows that the effect of front-end enhancement is model dependent. A preprocessing method that is useful for one internal representation may have a smaller effect on another.

The difference is also useful for system design. If an application uses features from an early layer, Wiener enhancement may provide a clear benefit. If it uses a later representation, the benefit may be much smaller.

5.5 Waveform Quality Only Partly Predicts Representation Quality

The relationship between SNR improvement and representation recovery was positive, especially for WavLM, but it was not one-to-one.

Some conditions with similar SNR gains produced noticeably different representation recovery. This means that waveform improvement contains useful information about model behaviour, but cannot completely predict it.

This result is important when speech enhancement is used before a neural speech model. Evaluating only the enhanced waveform may miss effects that become visible inside the downstream model.

For systems intended for machine processing rather than only human listening, representation-level evaluation is therefore useful in addition to conventional enhancement metrics.

5.6 Limitations and Future Work

The experiment used four clean utterances from two speakers, so the results describe a relatively small controlled dataset. A larger set of speakers and utterances would provide more stable averages.

Only DKITCHEN and STRAFFIC were tested. Adding more DEMAND environments would show whether the same trends appear for other types of stationary and non-stationary noise.

The Wiener system used an initial noise-only segment for noise estimation. A practical real-time system could instead use adaptive noise tracking or voice activity detection.

Only layers 1, 4, 8 and 12 were analysed. Evaluating every layer could show more precisely where the representation benefit begins to decrease.

Finally, this project measured representation similarity rather than a specific downstream task. A useful extension would be to test whether the same conditions improve automatic speech recognition or another speech-processing task.

6 Conclusion

This project studied whether classical Wiener filtering helps modern speech foundation models, rather than evaluating enhancement only at the waveform level.

Four LibriSpeech utterances were mixed with two real DEMAND noise environments at four SNR levels, producing 32 controlled experiments. The same Wiener system was evaluated using SNR, STOI, and layer-wise representations from HuBERT Base and WavLM Base+.

Wiener filtering consistently improved waveform SNR. The average improvement was approximately 7.27 dB for DKITCHEN and 4.64 dB for STRAFFIC. STOI, however, changed only slightly.

The representation results showed a stronger layer-dependent pattern. Early HuBERT and WavLM representations generally became more similar to clean speech after Wiener enhancement. The benefit decreased with network depth, especially for HuBERT, where mean recovery dropped from 0.0551 at layer 1 to 0.0043 at layer 12. WavLM retained a larger final layer recovery of 0.0178.

The research question can therefore be answered as follows:

> **Wiener filtering helps modern speech foundation models most clearly at early acoustic layers and under strongly noisy conditions. Improving waveform SNR does not guarantee the same improvement in deeper neural representations.**

The project shows that speech enhancement should be evaluated not only with waveform metrics, but also in the representation space of the model that will ultimately use the processed speech.

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